

# Mooring Design Study

Soft moorings for the single buoy, the 4-buoy cluster and the 4×4 platform: criteria, selection, verification and installation checks

# Specification complete; interference and excursion met

|                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Status</p> <p><b>Ready, one decision open</b></p> <p>Rev D is complete. Procurement is gated on HWRL confirming facility assumptions F1–F7 (e.g. underwater wall anchors rated <math>\geq 300</math> N) and on one decision: the 2.4 m platform's 0.49 m wall clearance.</p> | <p>Criterion 6</p> <p><b>Interference within <math>\zeta/3</math></b></p> <p>Tilt natural period shifts <math>-1.37 / -0.73 / -0.74</math> % (buoy / cluster / platform) against <math>\zeta/3 = 1.42 / 1.19 / 1.01</math> %. Heave natural period shifts <math>\leq 0.50</math> %.</p> | <p>Criterion 4</p> <p><b>Excursion <math>\leq 1.0</math> m</b></p> <p>At <math>H = 0.5</math> m the extreme cord set (<math>k \times 5 / \times 5.5</math>) holds the max surge to <math>0.97 / 0.94</math> m. The operational set would drift to <math>3.39 / 3.51</math> m.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

One anchor layout, two cord sets: **operational** ( $H \leq 0.12$  m, minimal interference) and **extreme** ( $H = 0.2\text{--}0.5$  m, station-keeping).

# Three articles, natural periods inside the wave band

| Article      | Configuration                   | Mooring points         | Heave T <sub>n</sub> | Tilt T <sub>n</sub> |
|--------------|---------------------------------|------------------------|----------------------|---------------------|
| Single buoy  | 1 spar with a heave plate       | radial collar, 4 lines | 2.561 s              | 2.761 s             |
| Cluster      | 4 buoys hinged to a hub, at 45° | 4 spars, 4 lines       | 2.586 s              | 2.856 s             |
| 4×4 platform | 16 buoys at 45°, 2.69 m wide    | 8 row spars, 4 bridles | 2.599 s              | 2.926 s             |

| Band        | Wave height H      | Wave period T (regular, heading 0)           | Purpose                        |
|-------------|--------------------|----------------------------------------------|--------------------------------|
| Operational | 0.04, 0.08, 0.12 m | 1.4–3.5 s; 0.05 s step across the resonances | motions: FloatSim validation   |
| Extreme     | 0.2, 0.35, 0.5 m   | a subset, steepness $H/\lambda \leq 0.08$    | line and anchor loads, offsets |

**Mesh correction: these T<sub>n</sub> are long**

T<sub>n</sub>: unmoored, FloatSim modal analysis on the flume BEM. Its 12-sided waterline reads C33 and C55 4.5 % low, so every T<sub>n</sub> above is long by ~0.04 s (heave) and ~0.06–0.08 s (tilt). Converged mesh, single buoy: **heave 2.519 s, pitch 2.684 s**, matching the OSU buoy model (2.52 / 2.69 s) and the field heave-decay video (≈ 2.50 s). The mooring's shifts are relative and unaffected.

# Station-keeping and non-interference conflict

## Station-keeping

- max surge, mean + dynamic,  $\leq 1.0$  m: tracking window  $-0.3 \dots +1.0$  m
- $|\text{mean sway}| \leq 0.10$  m; wall clearance  $\geq 0.6$  m
- no slack in the operational band:  $T_{\min} \geq 0.15 T_0$
- $WLL \geq 3 \times \text{max tension}$ ; stroke  $\geq 1.25 \times \text{max stretch}$

## Non-interference

- tilt natural period shift  $\leq \zeta/3$ : 1.42 / 1.19 / 1.01 %
- heave natural period shift  $\leq 1$  %
- calm static tilt  $\leq 1^\circ$
- surge, sway, yaw periods  $\geq 4 \times 3.5 \text{ s} = 14 \text{ s}$

Stiff lines hold station; soft lines minimise interference.  $\delta \leq \zeta/3$  keeps the resonant amplitude  $\geq 1/\sqrt{(1 + 1/9)} = 94.9$  % of the unmoored peak.

# Criteria first, every option simulated in FloatSim

- 1 Criteria agreed before design.** Seven pass/fail criteria, confirmed in Phase B.
- 2 FloatSim for every option.** Moored decks, static settle, modal analysis, rigid pulls, and regular-wave runs with wave-relative Morison drag and the drift sum applied in-run.
- 3 Selection by a written rule.** Hard criteria, then the most pretension, the attachment closest to the waterline, the stiffest surge.
- 4 Verification runs.** The chosen designs in FloatSim wave runs, operational band and  $H = 0.5$  m.
- 5 Generated documents.** The specification's tables and figures are built from the same JSON records.

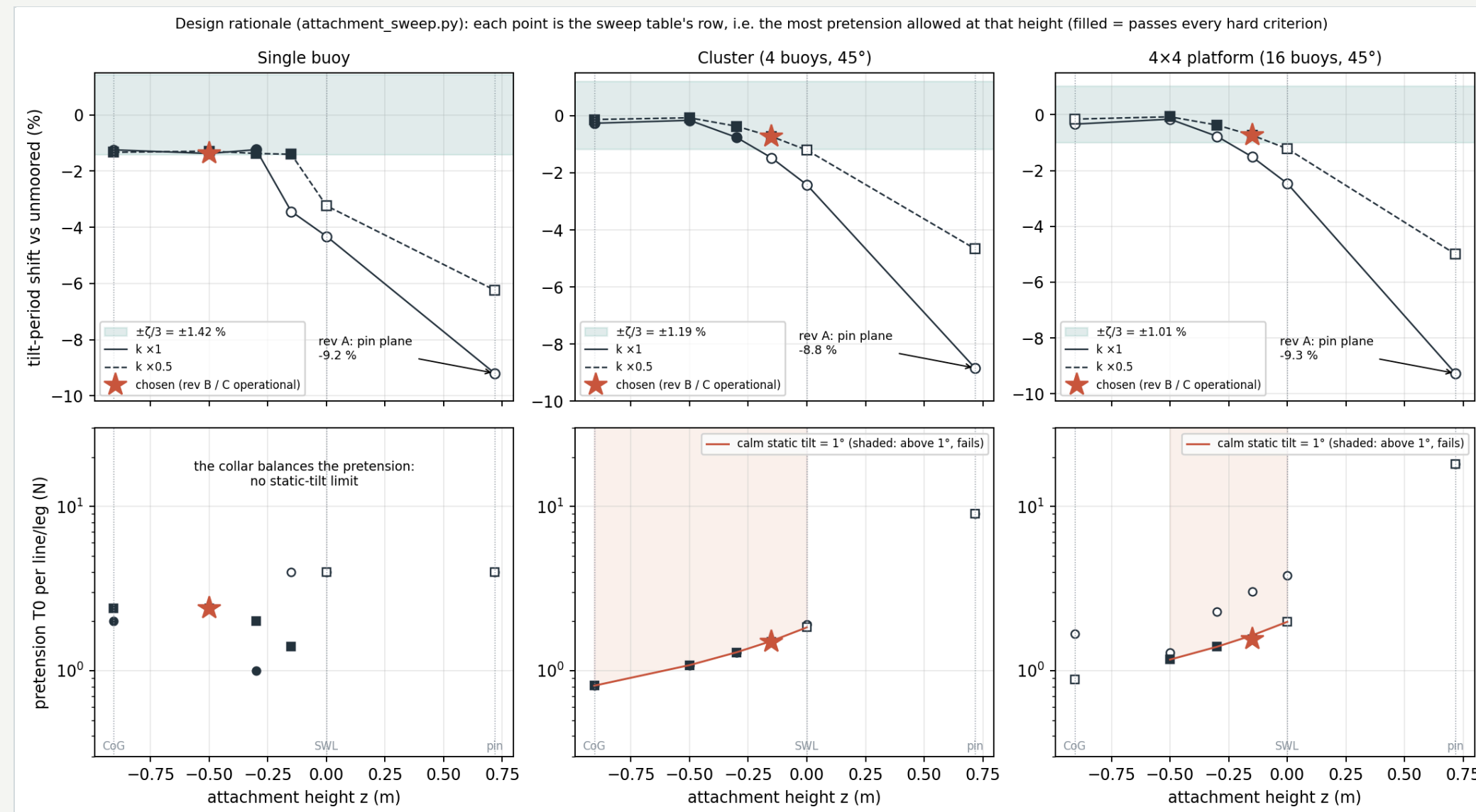
Drift sum: the recorded fixed-body bound (splash-zone drag + Havelock term) plus FloatSim's own mean force, so conservative. LEVEL1 kinematics are valid to  $5.7^\circ$ : resonant results are indicative.

# Five revisions, each correcting the last

| Rev | Design                                                                 | Finding                                                                                |
|-----|------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| A   | Lines at the pin plane (+0.717 m)                                      | Tilt T_n -9.19 / -8.83 / -9.04 %, 6.5–8.6 × ζ/3: withdrawn                             |
| B   | Attachment sweep: z -0.50 / -0.15 / -0.15 m, low T0                    | Within ζ/3, but H = 0.5 m drifts 2.5–3.4 m: criterion 4 was missing from the rule      |
| C   | Rev B as the operational set + an extreme set, k ×5 / ×6, same anchors | Max surge 0.97 / 0.95 m at H = 0.5 m; its surge periods 10.5 / 9.6 s declared          |
| C.1 | Creep allowance, ~4 % of the stretch per decade (assumed)              | Creep table, daily re-check and re-tension thresholds                                  |
| D   | Platform resized to the test matrix's 2.4 m square, re-derived         | Tilt -0.74 % (ζ/3 1.01 %); extreme ×5.5, max surge 0.94 m; wall clearance 0.49 m: open |

Phase H evaluated removing the pretension and kept it (no spec change): see the decisions.

# Attach below the waterline: coupling $\approx k \cdot h^2$



## Two reference points

The static pretension moment acts about the pin (+0.717 m). The dynamic coupling,  $\approx k \cdot h^2$ , acts about the tilt mode's rotation centre, near the CoG (-0.907 m).

Tilt T\_n shift, buoy / cluster / platform

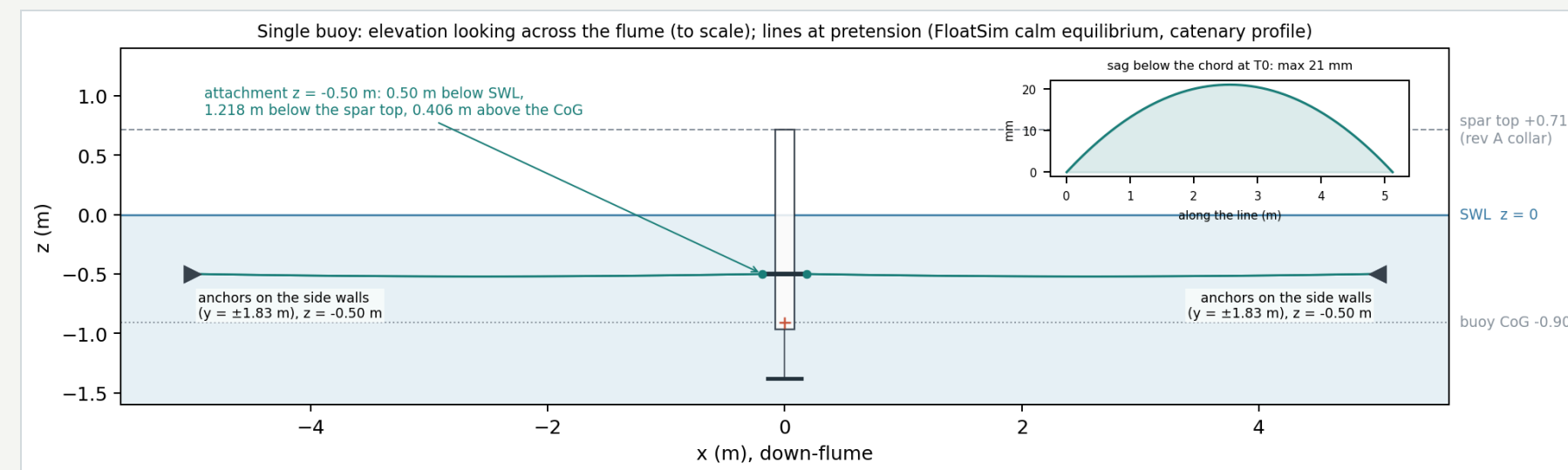
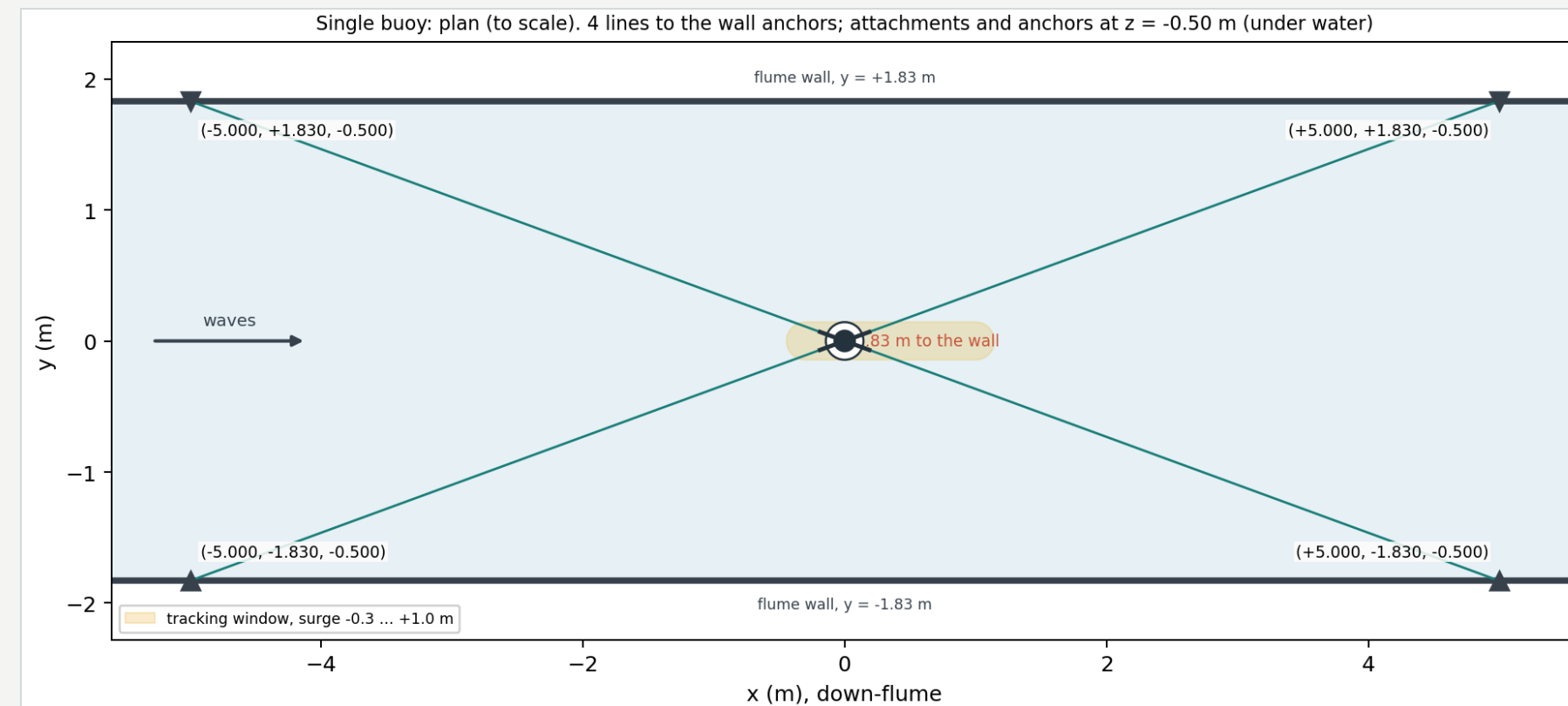
Pin plane: -9.2 / -8.8 / -9.3 %

Chosen: -1.37 / -0.73 / -0.74 %

Limit  $\zeta/3$ : 1.42 / 1.19 / 1.01 %

## Trade-off

Below the pin, the pretension tilts the cluster and platform:  $1.00^\circ$  calm tilt,  $\approx 0.70^\circ/\text{N}$ , used as the pretension gauge.



Configuration 1 of 3 · to scale

# Single buoy

Four lines from a radial collar to wall anchors at  $x = \pm 5.0$  m,  $y = \pm 1.83$  m. The collar balances the lines: calm tilt  $0^\circ$ .

## Attachment

radial collar  $r$  0.2 m,  $z$   $-0.50$  m

## Cord, one set

$k$  2.01 N/m,  $T_0$  2.40 N at rest

Heave  $T_n$ , free  $\rightarrow$  moored (flume BEM)

2.561  $\rightarrow$  2.548 s ( $-0.50\%$ )

Tilt  $T_n$ , free  $\rightarrow$  moored (flume BEM)

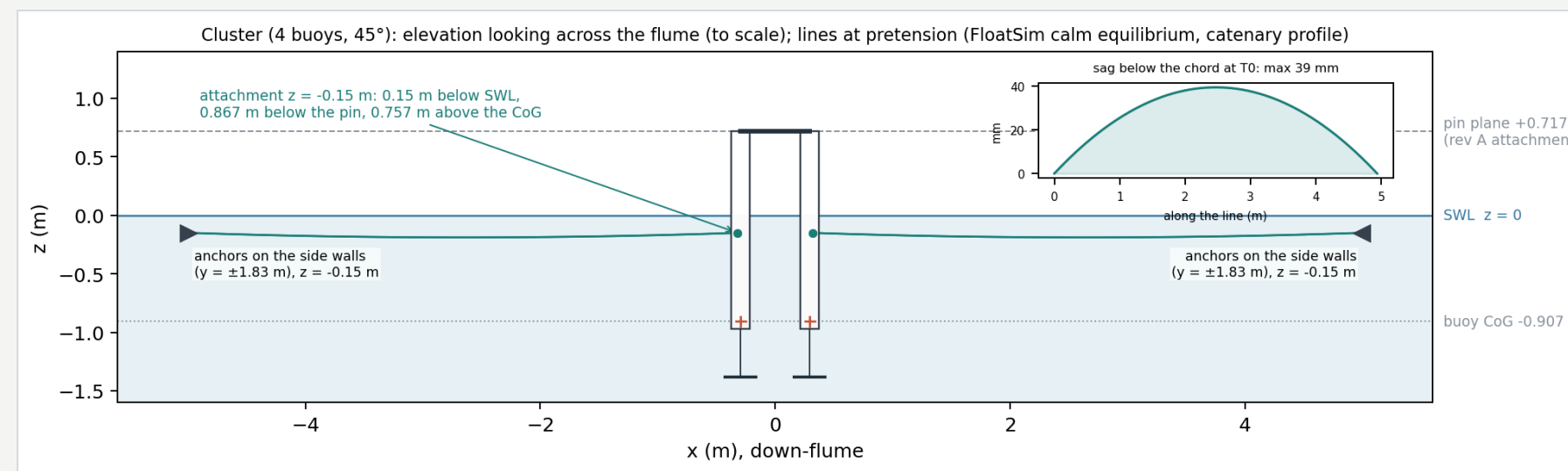
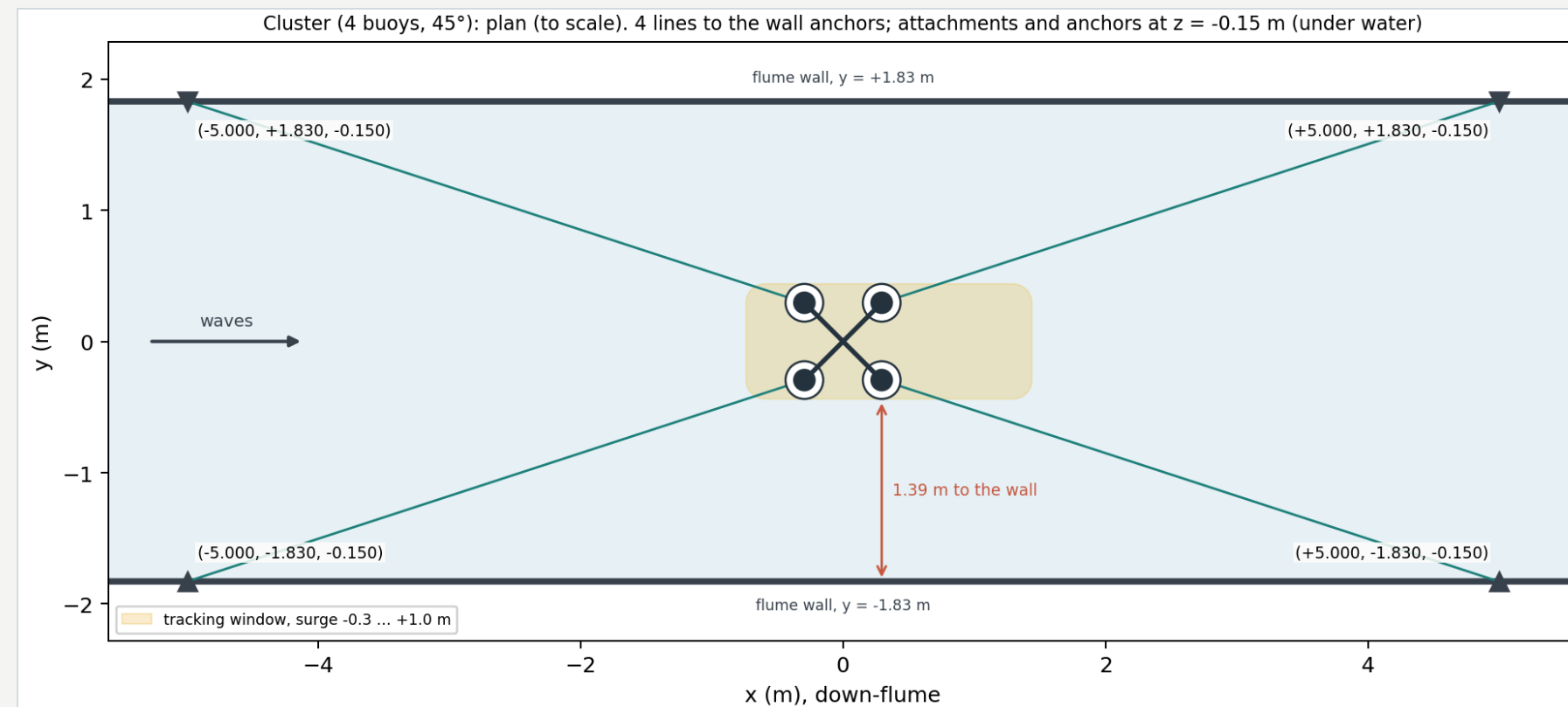
2.761  $\rightarrow$  2.723 s ( $-1.37\%$ )

Converged mesh, free: heave · pitch

2.519 s · 2.684 s (video  $\approx$  2.50 s)

$H = 0.5$  m max surge

0.68 m (predicted cases)



Configuration 2 of 3 · to scale

## Cluster

4 buoys hinged to a hub at 45°. One line per spar to wall anchors at  $x = \pm 5.0$  m,  $y = \pm 1.83$  m.

### Attachment

each spar,  $z -0.15$  m

### Cords, operational / extreme

$k$  4.15 /  $\geq 20.75$  N/m, 1.44 N at rest

### Heave $T_n$ , free $\rightarrow$ moored

2.586  $\rightarrow$  2.584 s (−0.08 %)

### Tilt $T_n$ , free $\rightarrow$ moored

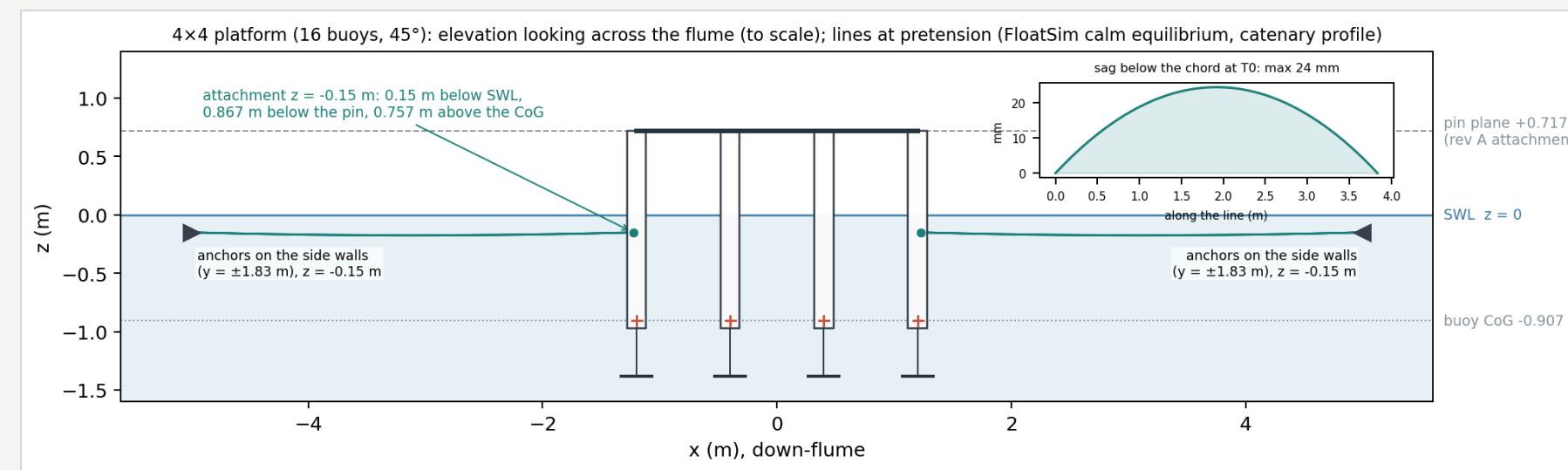
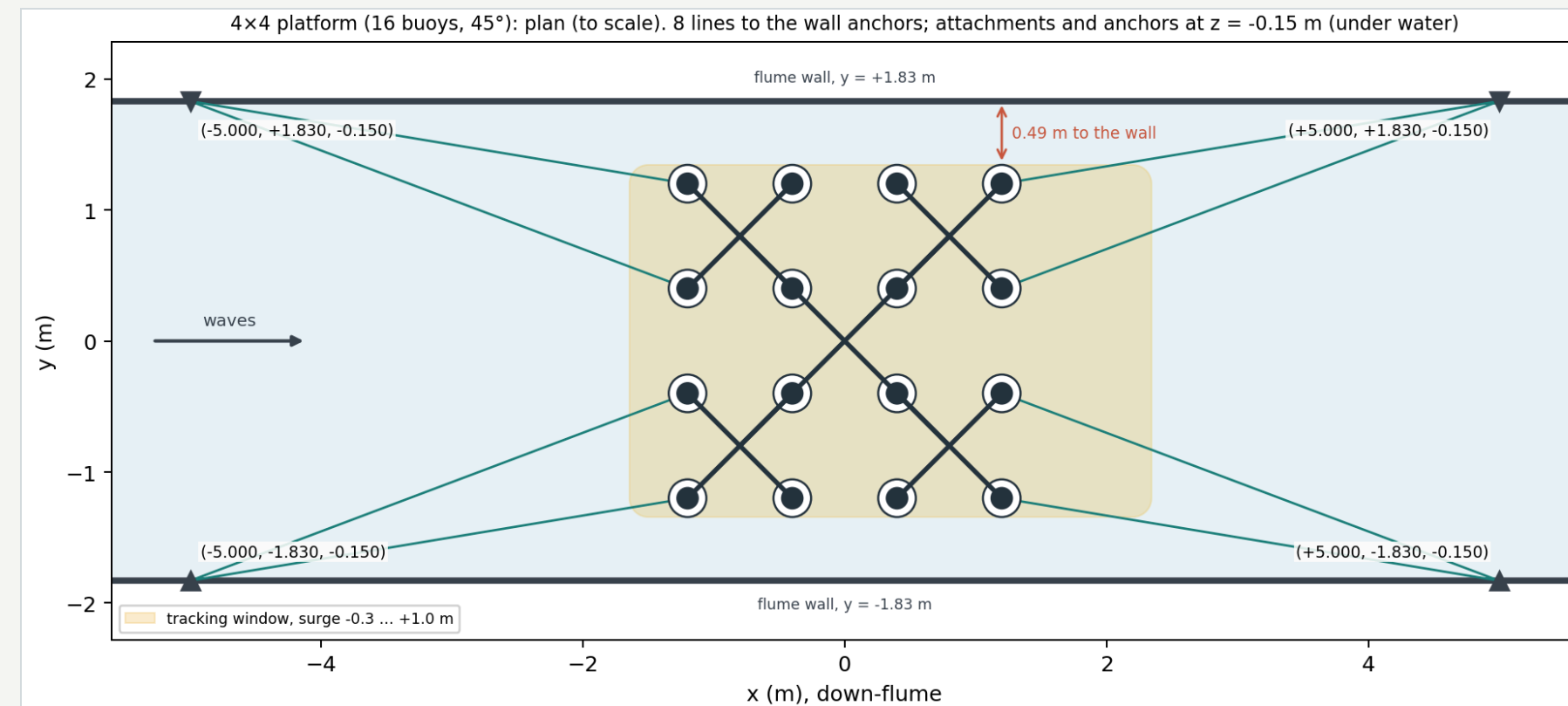
2.856  $\rightarrow$  2.835 s (−0.73 %)

### Calm tilt · wall clearance

1.00°, both sets · 1.39 m

$H = 0.5$  m max surge, extreme set

0.97 m



Configuration 3 of 3 · to scale

## 4x4 platform

16 buoys at 45° on a 2.4 m square, 2.69 m wide. Four 2-leg bridles to the row spars; anchors at  $x = \pm 5.0$  m,  $y = \pm 1.83$  m.

### Attachment

8 row spars,  $z = -0.15$  m

### Cords per leg, operational / extreme

k 8.36 /  $\geq 45.96$  N/m, 1.44 N at rest

### Heave $T_n$ , free $\rightarrow$ moored

2.599  $\rightarrow$  2.598 s (−0.04 %)

### Tilt $T_n$ , free $\rightarrow$ moored

2.926  $\rightarrow$  2.904 s (−0.74 %)

### Calm tilt · wall clearance

1.00°, both sets · 0.49 m (criterion  $\geq 0.6$  m)

H = 0.5 m max surge, extreme set

0.94 m

# The mooring barely moves the natural periods

| Article      | Tilt T <sub>n</sub> shift (limit ζ/3) | Heave T <sub>n</sub> shift (limit 1 %) | Heave RAO Δ vs free at the tilt resonance, H 0.04 / 0.12 m | Calm tilt |
|--------------|---------------------------------------|----------------------------------------|------------------------------------------------------------|-----------|
| Single buoy  | −1.37 % (1.42)                        | −0.50 %                                | −0.2 / +2.5 %                                              | 0°        |
| Cluster      | −0.73 % (1.19)                        | −0.08 %                                | +3.2 / +1.9 %                                              | 1.00°     |
| 4×4 platform | −0.74 % (1.01)                        | −0.04 %                                | +5.9 / +3.4 %                                              | 1.00°     |

### Heave

The natural period moves ≤ 0.50 %, about 1/7 of the heave ζ/3 (ζ ≈ 11 %). The largest heave RAO change, +9.0 % (buoy) and +7.8 % (cluster) at H = 0.12 m, is at T = 2.2 s: off resonance, inside the slosh exclusion.

### Tilt

Peak tilt, moored – free: +0.79 to +1.10° off resonance, i.e. the 1° calm tilt; +0.46 / +0.57° at the cluster and platform resonance; buoy ≤ +0.39°.

### Open: Phase D

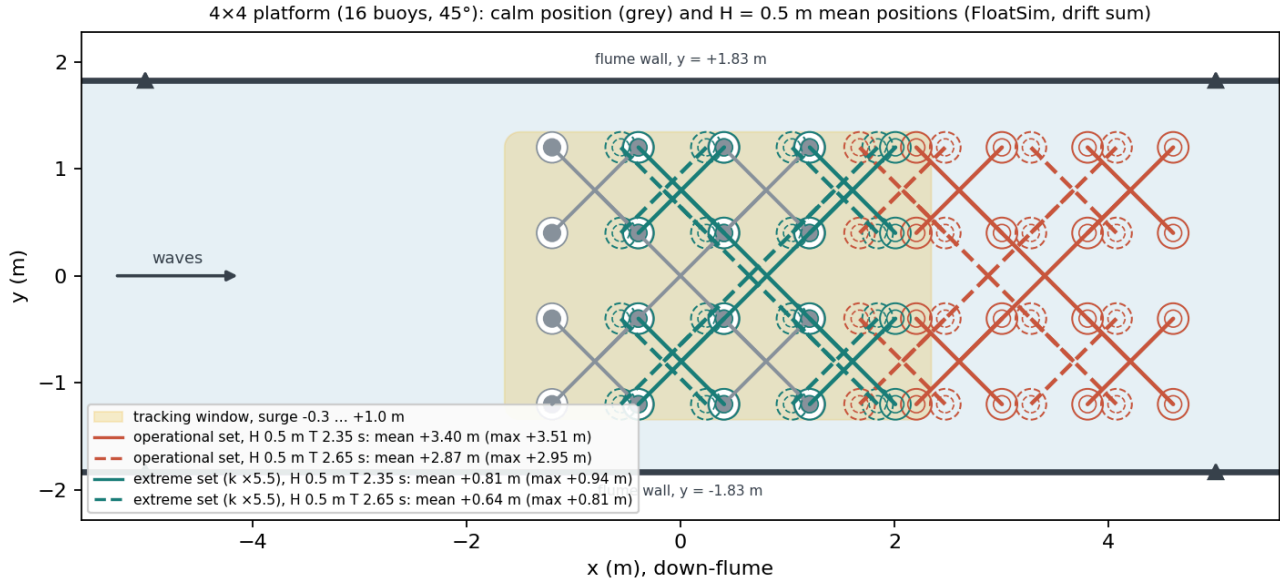
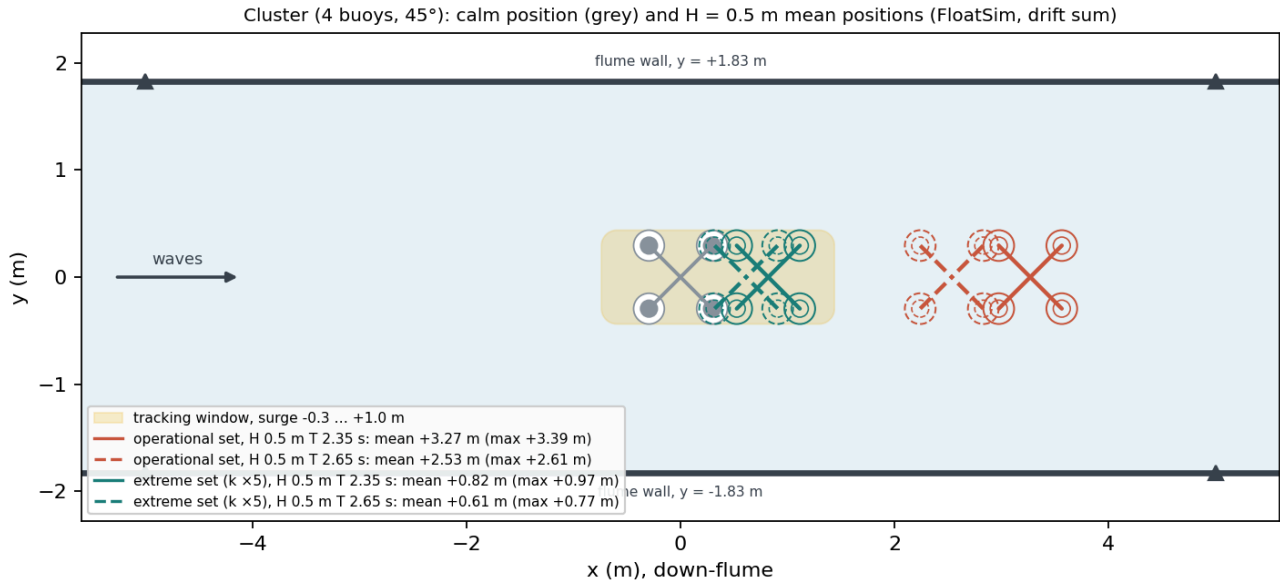
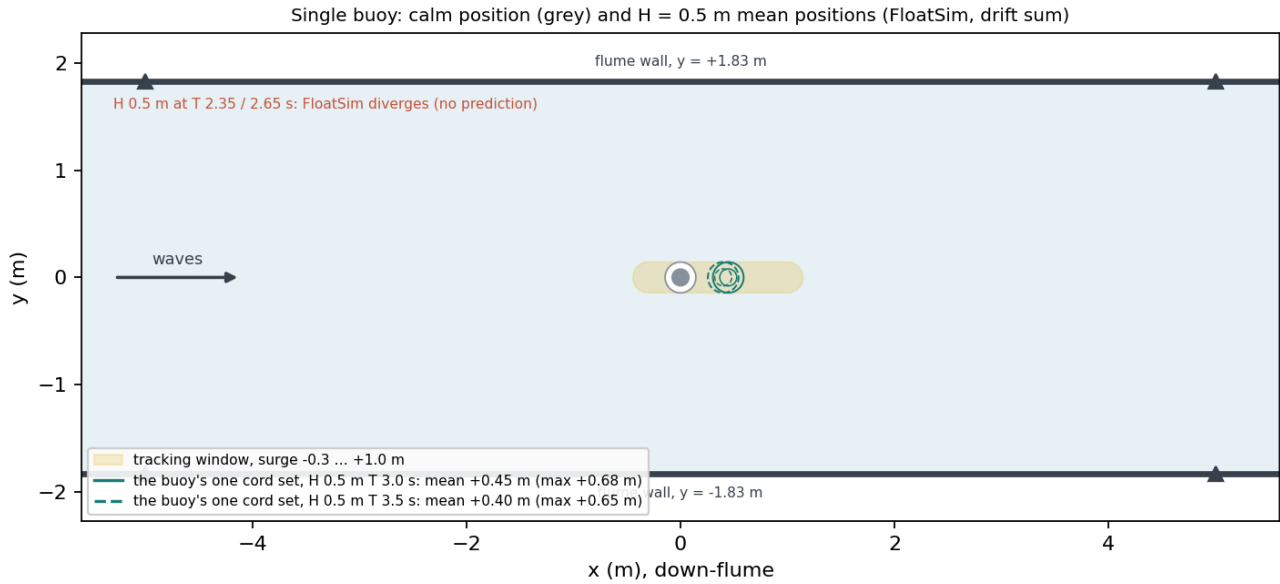
Criterion 6 asks ±3 % on the heave RAO; +3.2 % (cluster), +5.9 / +3.4 % (platform) and the 2.2 s values exceed it. The formal check, with restrained free runs, is Phase D.

Result 2 · excursion at H = 0.5 m, FloatSim, drift sum

# The extreme set keeps H = 0.5 m in view

| Article  | Cord set     | T 2.35 s: mean / max | T 2.65 s: mean / max |
|----------|--------------|----------------------|----------------------|
| Cluster  | operational  | 3.27 / 3.39 m        | 2.53 / 2.61 m        |
| Cluster  | extreme ×5   | 0.82 / 0.97 m        | 0.61 / 0.77 m        |
| Platform | operational  | 3.40 / 3.51 m        | 2.87 / 2.95 m        |
| Platform | extreme ×5.5 | 0.81 / 0.94 m        | 0.64 / 0.81 m        |

- Operational band ( $H \leq 0.12$  m): mean offset  $\leq 0.07$  m, no slack ( $T_{\min} / T_0 \geq 0.18$ ).
- At  $H = 0.5$  m the down-flume lines go slack and stay slack; the up-flume lines carry the drift (peak 22.1 / 42.0 N).
- Single buoy: one set; its predicted  $H = 0.5$  m cases reach 0.68 m.



# One anchor layout, two cord sets

| Cluster / platform          | Operational set                 | Extreme set                                 |
|-----------------------------|---------------------------------|---------------------------------------------|
| Use                         | response tests, $H \leq 0.12$ m | loads and survival, $H = 0.2\text{--}0.5$ m |
| Cord stiffness $k$          | 4.15 / 8.36 N/m (−13 / +30 %)   | 20.75 / 45.96 N/m (−0 / +30 %)              |
| At-rest tension · calm tilt | 1.44 / 1.44 N · 1.00°           | the same (at-rest-matched) · 1.00°          |
| Pre-stretch at rest         | 346 / 172 mm                    | 69 / 31 mm: set by tension, not length      |
| Surge natural period        | 23.1 / 22.9 s                   | 10.5 / 9.9 s (criterion 2 declared)         |
| Surge pull stiffness $K$    | 15.0 / 61.7 N/m                 | 73.2 / 330.0 N/m                            |

**Identification:** the calm tilt cannot tell the sets apart, so a static pull is required before every extreme series: the extreme set reads  $K \geq 33$  N/m (cluster) or  $\geq 143$  N/m (platform).

# Key decisions and their technical basis

| Decision                          | Basis                                                                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keep the operational pretension   | Taut lines in every operational case. Without it, the lines lose stretch for 18–41 % of each resonant cycle, which the quasi-static line cannot model. |
| Keep the buoy's pretension        | The collar turns T0 into yaw stiffness; at the taut minimum the yaw period would be 2.48 s, inside the wave band.                                      |
| Two cord sets, not one compromise | Criteria 2 and 4 cannot both hold at H = 0.5 m: surge $\geq 14$ s caps k at $\sim \times 2.7\text{--}2.8$ , leaving $\geq 1.5$ m of offset.            |
| Size the extreme set on max surge | Criterion 4 is mean + dynamic $\leq 1.0$ m; the mean alone would allow $\times 4.5$ / $\times 5$ .                                                     |
| Match the at-rest tension         | Identical calm line forces: static residual 0.0073 / 0.0087 N, against 0.13 / 0.29 N at the same nominal T0.                                           |

# Anchors, cords, collar and the daily re-check

## Wall anchors

Four per article at the attachment depth, under water, each rated  $\geq 300$  N. The largest working load ( $\times 3$  on the FloatSim peak) is 246.9 N: platform, up-flume, extreme set.

## Elastic cords

Usable submerged,  $\approx 0.02$  N/m. Operational stiffness  $-13 / +30$  %; extreme stiffness a minimum,  $-0 / +30$  %. Elongation capacity  $\times 1.25$  on the peak stretch.

## Buoy collar

A slender spreader, round section  $\leq 25$  mm (added-mass budget 0.4 kg). A solid disk of  $r = 0.2$  m would add  $\approx 21$  kg of heave added mass: rejected.

## Creep: daily re-check

$\sim 4$  % per decade assumed: 87.4 % of the at-rest tension is left after a day (calm tilt  $1.00 \rightarrow 0.87^\circ$ ). Re-tension below the thresholds; the platform (operational) and the buoy before their critical cases.

# Risks and limitations

| Item                                                   | Handling                                                                                                         |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| HWRL facility assumptions F1–F7 unconfirmed            | Confirm before procurement; any “no” forces a redesign                                                           |
| Platform wall clearance 0.49 m, criterion $\geq 0.6$ m | Sidewall effect $\leq 3.3$ %, no larger than at 0.80 m: accept on that evidence, or reduce the platform (Xabier) |
| LEVEL1 valid to 5.7°; resonant tilts 10–31°            | Resonant results indicative; LEVEL2 rerun of the resonant subset                                                 |
| Buoy extremes: 3 cases diverge, 6 grow yaw             | Run last with a height ramp, stop on yaw growth; pitch-neutral collar                                            |
| Heave RAO change above $\pm 3$ % in some cases         | Formal criterion-6 check with restrained free runs in Phase D                                                    |
| Snap loads and the return swing not simulated          | $\times 3$ load factor; load-cell pull test at installation                                                      |
| Flume BEM waterline 12-sided (NT12)                    | Resonances $\sim 0.04\text{--}0.08$ s long; NT36 regeneration pending                                            |

Next steps

## From specification to the flume

- 1** **HWRL confirmation of F1–F7** and the platform's 0.49 m wall clearance decision. Underwater wall anchors, the  $\geq 300$  N rating, cord in two stiffnesses: gates procurement.
- 2** **Procurement.** Both cord sets to the line tables; creep rate from the datasheet or a one-day hold test.
- 3** **Verification.** BEM at NT36 and the Phase D matrix, re-planned on the HSFP test matrix: both cord sets, criterion 6 included.
- 4** **LEVEL2.** Rerun of the resonant subset: tilts, loads, offsets and the buoy's divergence.

# Where every number comes from

| Record                                       | Holds                                                         |
|----------------------------------------------|---------------------------------------------------------------|
| MOORING-SPEC.md / .pdf, rev D                | the specification: line tables, figures, checks, procedure    |
| DESIGN-BASIS.md                              | every decision and its evidence, Phases A–J                   |
| attachment_sweep.py → .json                  | the operational set: the sweep, the rule, the natural periods |
| extreme_set.py → .json                       | the extreme set: k scan, final runs, residuals                |
| spec_statics.json, spec_statics_extreme.json | statics: anchors, attachments, T0, pull K, periods            |
| pretension_study.json, PRETENSION-EVAL.md    | the pretension evaluation                                     |
| viewer_rows_revc/ (48 runs)                  | moored vs free comparison; the 3D motion viewer               |
| mooring_figures.py → figs/                   | the plan, elevation, rationale and offsets figures            |